ETH zürich | Physikalisches Praktikum der ETH Zürich | Physics Lab

Name 1 Datum:

Name 2 Platz-Nr:

60 - Basics in Electronics

Part II - Measurements with the Multimeter

1. Internal Resistance of the Voltmeter

- Estimation (homework!) for $R_1 = 4.7\text{k}\Omega, R_2 = 10\text{k}\Omega$. $U_{\text{out}} =$_____
- Estimation (homework!) for $R_1 = 4.7\text{M}\Omega, R_2 = 10\text{M}\Omega$. $U_{\text{out}} =$_____
- Measurement for $R_1 = 4.7\text{k}\Omega, R_2 = 10\text{k}\Omega$. $U_{\text{out}}^{\text{meas}} =$_____
- Measurement for $R_1 = 4.7\text{M}\Omega, R_2 = 10\text{M}\Omega$. $U_{\text{out}}^{\text{meas}} =$_____

Since the measured and estimated voltages doe not match, the internal resistance cannot be neglected! But it can be calculated to be: Internal resistance of the multimeter $R_i =$_____

2. Compensation Circuit

If $U_{\text{Diff}} = 0$, then we read $U_{\text{Pot}} =$_____ What does it mean? What's the benefit of this method?

3. Internal Resistance of the Amperemeter

- Estimation (homework!): $I_1 =$_____, $I_2 =$_____, $I_3 =$_____
- Measurement: $I_1^{\text{meas}} =$_____, $I_2^{\text{meas}} =$_____

And thus we can calculate the internal resistance of the amperemeter to be $R_i =$_____

Is this expected? Why? _____

_____

Why is R_i for the measurement in the µA-range so high? _____

_____

4. Four-Probe Measurement (optional, except for D-PHYS!)

Diameter of the copper wire: _____ Length of the copper wire: _____

ρ_{Cu} = _____ Literature value: $\rho_{\mathrm{Cu}}^{\mathrm{lit}}$ = _____

Diameter of the unknown wire: _____ Length of the unknown wire: _____

ρ_{unknown} = _____ Guess what material it is: _____

Part IV – Measurements with the Oscilloscope

4. Four-Probe Measurement (optional, except for D-PHYS!)

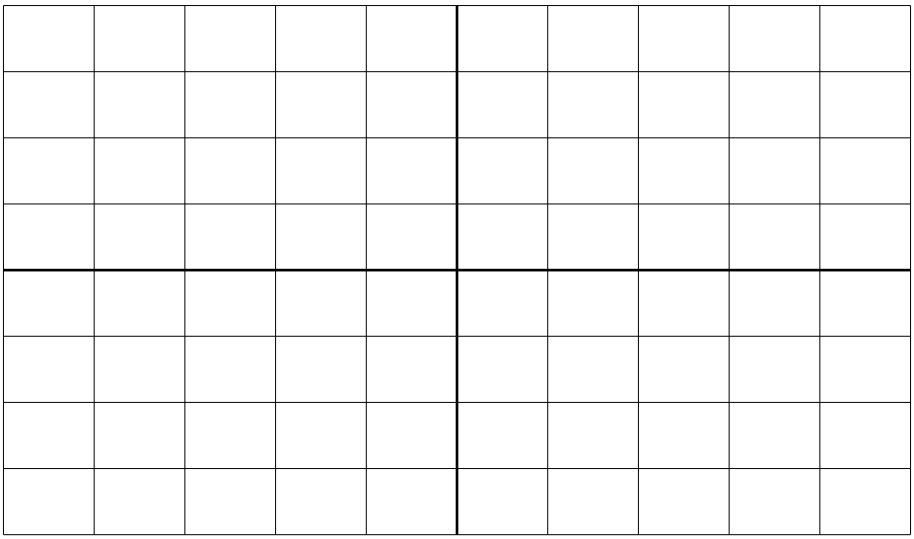
1. Measurement of the PROBE-COMP signal

$V_{\text{peak-peak}}$=_____ f =_____ mean =_____

2. Multi Channel Measurement on Diodes

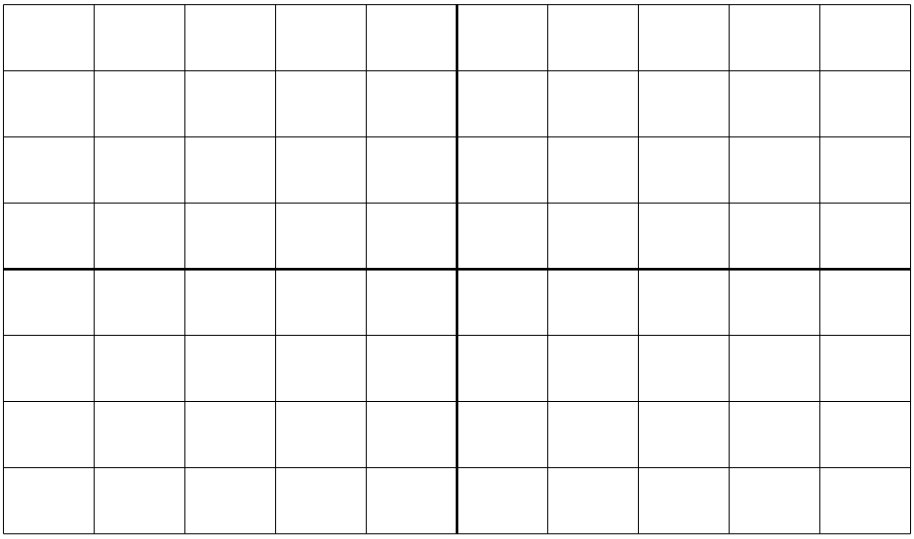
V_{forward} = _____

3. Difference Measurement



Oscilloscope Screen showing the difference signal

4. Using a Capacitor to rectify the Signal (at different frequencies)



Rectifier

Diode is conducting during a time t_{cond} =_____

Raising slope: _____

Falling slope: _____

Ripple voltage at $U_{\text{peak-peak}}$ at 100 Hz = _____

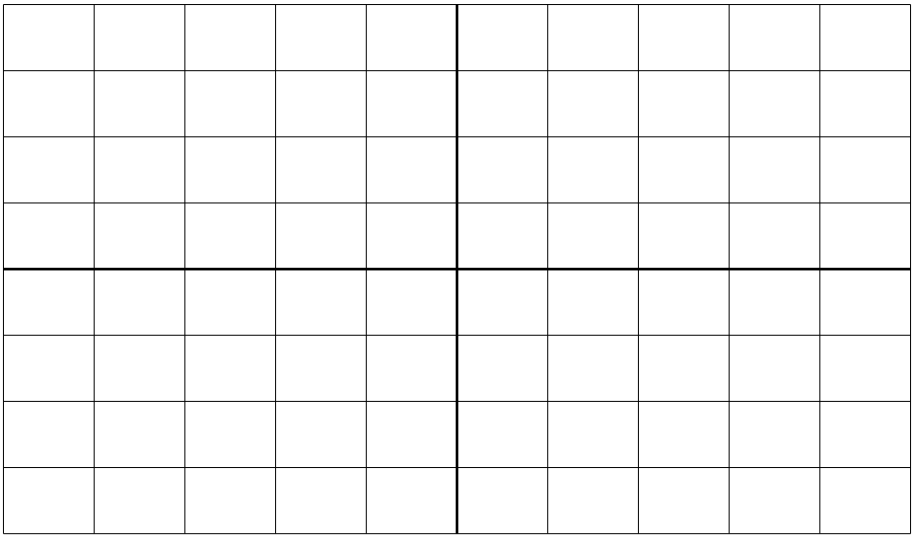
Ripple voltage at $U_{\text{peak-peak}}$ at _____ Hz = _____

Ripple voltage at $U_{\text{peak-peak}}$ at _____ Hz = _____

Explanation? _____

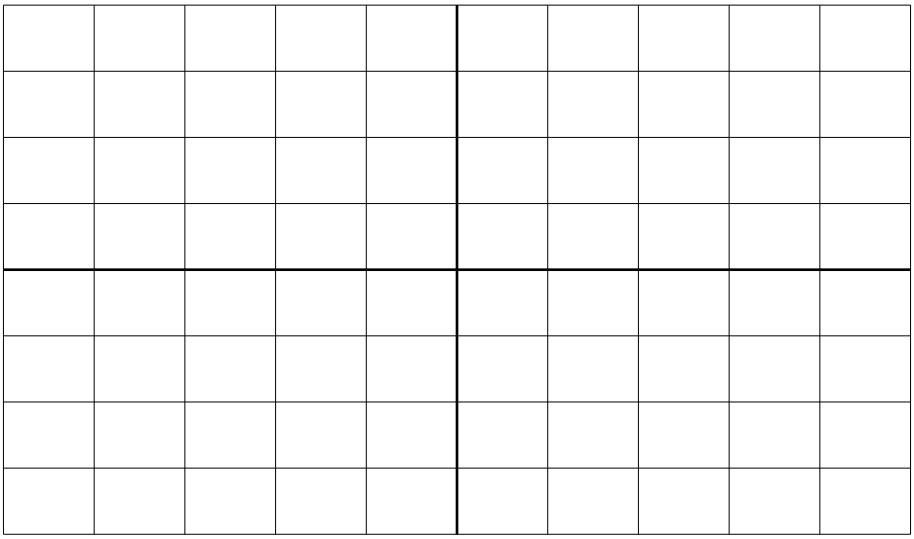
_____

4. Behaviour of Diodes at Higher Frequencies



Behaviour at 100 kHz

5. Single-Shot Measurements



Bounced and debounced signal

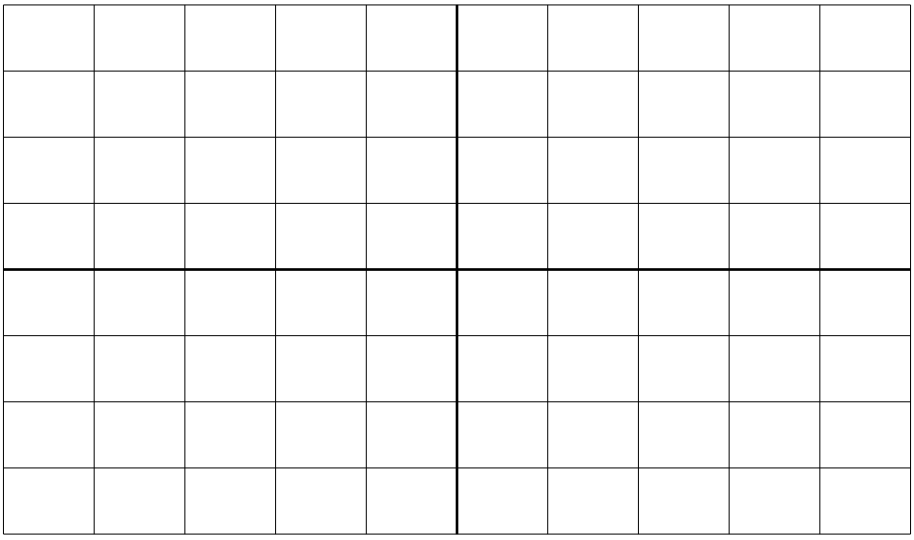
6. Low-pass filter (optional, except for D-PHYS)

Measurement 1: _____

Measurement 2: _____

Measurement 3: _____

7. (if time permits) X-Y-Mode on the Oscilloscope



Characteristic Curve of Diodes